MIL-S-46997(MI)
23 JULY 1963
SUPERSEDING
MPD 5016
29 NOVEMBER 1960

MILITARY SPECIFICATION

SQUIB, ELECTRIC, M3, MANUFACTURE AND ASSEMBLY

1. SCOPE

1.1 This specification covers the manufacture and assembly of one type of electric squib.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, (or request for proposal), form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-B-838 — Butyl Acetate: Normal (For use in Organic Coatings)

PPP-B-566 — Boxes, Folding, Paperboard

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 — Box, Fiberboard

PPP-C-843 — Cushioning Material, Cellulosic

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

MILITARY

MIL-W-76 — Wire and Cable, Hook Up, Electrical Insulated

MIL-P-116 — Preservation, Methods of

MIL-B-131 — Barrier Material; Water Vaporproof, Flexible

JAN-Z-399 — Zirconium, Powdered (For Use in Ammunition)

JAN-S-732 — Shellac Solutions (For Use in Ammunition)

MIL-A-2550 — Ammunition and Special Weapons, General Specification for

MIL-W-5086 — Wire, Electrical, 600-Volt, Copper, Aircraft

STANDARDS

FEDERAL

FED-STD-406 — Plastics: Methods of Test

FSC 1375

701044—63—(521)

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MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 — Marking for Shipment and Storage

MIL-STD-350 — Jolt Test for Use in Production of Fuzes

DRAWINGS

ORDNANCE CORPS

8030859 — Squib, Electric: M3

8030860 — Charge, Main Assembly

8030861 — Case

8030862 — Insulation Liner Disc

8030863 — Spacer

8030864 — Initiation Assembly

8030865 — Wire, Bridge, Assembly

8030866 — Plug Assembly

8030867 — Wire, Lead

8030868 — Wire, Bridge

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids (or request for proposal) shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 — Interstate Commerce Commission, Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Copies of the above document may be obtained from the superintendent of documents, Government Printing Office, Washington 25, D. C. Orders for copies should cite "*49 CFR 71-90 (Revised 1956)*").

3. REQUIREMENTS

3.1 Pilot lot. Unless otherwise specified (see 6.2), a preproduction sample shall be submitted prior to start of regular production and shall meet all the requirements of this specification.

3.2 Construction. The squibs shall conform to Drawing 8030859, and all drawings and specifications referenced thereon, and as specified herein (see 6.3.15 thru 6.3.19 and 6.4).

3.2.1 *Details of components.* The components of the squib shall be in two categories, inert and explosive.

3.2.1.1 *Inert components.* The inert components consist of the following assemblies and parts:

Nomenclature	*Drawing Number*
Case	8030861
Insulation Liner Disc	8030862
Spacer	8030863
Wire, Bridge, Assembly	8030865
Plug Assembly	8030866
Wire Lead	8030867
Wire, Bridge	8030868

3.2.1.1.1 *Plug assembly.*

3.2.1.1.1.1 *Dielectric characteristics.* The plug, prior to connecting the bridgewire, shall be capable of withstanding 1000 volts alternating current (ac) at ambient temperature for a minimum period of one second (sec.). The resistance of the plug between the lead wires shall be not less than 50 megohms (see 4.4.1.4).

3.2.1.1.1.2 *Cure.* The plug assembly shall not exceed an acetone extraction value of 5 percent when tested in accordance with 4.4.1.1.

3.2.1.1.1.3 *Leakage.* The plug assembly shall show no leakage when tested in accordance with 4.4.1.2.

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3.2.1.1.1.4 *Plug and lead wire tightness.* Individual lead wires shall withstand a pull test of 10 pounds without visible loosening in the plug when tested as specified in 4.4.1.3.

3.2.1.1.2 *Bridgewire strength.* Each bridgewire assembly shall withstand the loading test described in 4.4.2 as a routine manufacturing operation.

3.2.1.1.3 *Lead wire.* The lead wire shall be in accordance with Drawing 8030867 and designations LW–C22 (1) U7 and LW–C22 (1) U9 of MIL–W–76 except as follows:

(a) The wire shall be tinned in accordance with MIL–W–5086.

(b) The wire shall be insulated with approximately 0.006 inch of nylon. The nylon shall be Zytol Number (No.) 33 as made by E. I. DuPont deNemours, Wilmington, Delaware or equal.

3.2.1.2 *Explosive components.* The explosive components shall be in accordance with Drawings 8030860 and 8030864 and as specified herein.

3.2.1.2.1 *Initiation charge.*

3.2.1.2.1.1 *Composition.* The composition of the initiation charge mixture shall be as specified below:

Ingredient	*Percent by weight*
Zirconium metal powder, type I	45 plus or minus 1
Potassium perchlorate, Granulation I	47 plus or minus 1
Graphite powder	1 plus or minus 0.2
Binder, type I	7 plus or minus 0.5

Warning: **The initiation charge is a hazardous mixture, and shall be kept moist with Normal butyl acetate, conforming to Specification TT–B–838, until loaded onto the bridgewire assembly.**

3.2.1.2.1.1.1 *Zirconium, type I.* Type I zirconium shall have an average particle diameter of 2 microns or less when measured with a Fisher Sub Sieve Sizer. The Zirconium shall be screened (wet with water) through a U. S. Standard No. 325 mesh sieve (see 6.6).

3.2.1.2.1.1.2 *Potassium perchlorate.* The chemical composition of the potassium perchlorate shall be as specified below, (see 6.3.2).

Ingredient	*Percent*
Sulfates, maximum (max)	0.001
Iron, Max	0.0005
Chlorate and Chlorides (or Cl), max	0.010
Heavy metals (as lead), max	0.0005
Insoluble matter, max	0.005
Purity, minimum (min)	99.7

3.2.1.2.1.1.2.1 *Granulation I.* A minimum of 95 percent shall pass through a U. S. Standard No. 325 mesh sieve and have an average particle diameter of 5 plus or minus 3 microns when measured with a Fisher Sub-Sieve Sizer.

3.2.1.2.1.1.2.2 *Granulation II.* A minimum of 95 percent shall pass through a U. S. Standard No. 200 mesh sieve and 50 plus or minus 10 percent shall be retained on a U. S. Standard No. 325 mesh sieve.

3.2.1.2.1.1.3 *Graphite powder.* Graphite powder shall be electrically conductive and of spectroscopic purity. Contaminants shall be present only in parts per million. The powder shall be of such size that 90 plus or minus 10 percent shall pass through a U. S. Standard No. 325 mesh sieve (see 6.3.3).

3.2.1.2.1.1.4 *Binder, type I.* The binder, type I shall be a copolymer consisting of 87 percent vinyl chloride and 13 percent vinyl acetate (see 6.3.4).

3.2.1.2.1.2 *Loading initiation charge.* The bridgewire shall be coated with successive applications of the initiator mixture until a 0.012 plus or minus 0.003 gram bead has been formed (see 6.3.5 and 6.3.6).

3.2.1.2.1.3 *Conductive coating of initiation charge.* A conductive coating shall be applied to the dry initiation charge and the upper portion of the lead posts. The coating composition shall be as specified below (see 6.3.7).

Ingredient	*Percent*
Graphite powder (see 3.2.1.2.1.1.3)	80 plus or minus 2
Binder, type I (see 3.2.1.2.1.1.4)	20 plus or minus 2

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3.2.1.2.2 *Main charge.*

3.2.1.2.2.1 *Composition.* The composition of the main charge shall be as specified below:

Ingredient	*Percent by weight*
Zirconium, granular, type II	33 plus or minus 2
Aluminum, atomized	8.9 plus or minus 1
Aluminum, flake	3 plus or minus 0.5
Barium nitrate	50 plus or minus 2
Potassium perchlorate, granulation II	5 plus or minus 1
Binder, type II	0.1 plus or minus 0.05

***Warning:* The main charge shall be kept moist with 95 percent ethyl alcohol until loaded into the squib around the initiation charge.**

3.2.1.2.2.1.1 *Zirconium, type II.* Zirconium, type II, shall meet the detailed requirements of E–1, E–2, E–3, E–4 and E–5 of Specification JAN–Z–399 relating to chemical impurities, without exception, and shall be a homogeneous mixture of two granulations. The first granulation shall be 40 plus or minus five percent of the total mixture and shall be such that 100 percent passes through a U. S. Standard No. 140 mesh screen and 98 percent is retained on a U. S. Standard No. 200 mesh screen. The second granulation shall be 60 plus or minus five percent of the total mixture and shall be such that 100 percent passes through a U. S. Standard No. 200 mesh screen and is 100 percent retained on a U. S. Standard No. 325 mesh screen. Type II zirconium shall not contain any powder which passes through a U. S. Standard 325 mesh screen (see 6.3.8).

3.2.1.2.2.1.2 *Aluminum, atomized.* The atomized aluminum shall consist of spherical particles made from primary metal and shall contain a minimum of 98.5 percent free metallic aluminum. The aluminum shall be free of grease and shall be of such granulation that a minimum of 94 percent passes through a U. S. Standard No. 325 mesh sieve and has an average particle diameter of 13 plus or minus 3 microns when measured with a Fisher Sub-Sieve Sizer (see 6.3.9).

3.2.1.2.2.1.3 *Aluminum, flake.* The flake aluminum shall be a nonleafing type, unpolished, containing less than 1 percent grease and less than 4 percent oxide. The granulation shall be such that 90 percent passes through a U. S. Standard No. 325 mesh sieve (see 6.3.10).

3.2.1.2.2.1.4 *Barium nitrate.* The barium nitrate composition shall be as specified below:

Ingredient	*Percent*
Calcium (Ca)	0.05 max
Chloride (Cl)	0.0005 max
Heavy Metals (as Pb)	0.0005 max
Insolubles	0.010 max
Iron (Fe)	0.002 max
Purity	99.8 min
Strontium (Sr)	To Pass American Chemical Society Test (approximately 0.050 max)
Substances not precipitated by H_2SO_4 PH of 5 percent solution (25° C) (77° F)	5.0–8.0

The particle size shall be such that a minimum of 95 percent passes through a U. S. Standard No. 200 mesh sieve and 90 plus or minus 10 percent is retained on a U. S. Standard No. 325 mesh sieve (see 6.3.11).

3.2.1.2.2.1.5 *Binder, type II.* The type II binder shall be a copolymer consisting of 18.2 percent vinyl alcohol and 81.8 percent vinyl butyral (see 6.3.12).

3.2.1.2.2.2 *Main charge loading.* Approximately 90 milligrams of the main charge mixture shall be loaded into the spacer around the initiation charge (see 6.3.14).

3.2.1.2.2.3 *Main charge drying.* The main charge shall be dried in accordance with Drawing 8030860.

3.2.2 *Case coating.* The squib case shall be brushed or sprayed with shellac, conforming to type II of Specification JAN–S–732 to which a sufficient amount of basic fuchsin or brilliant green dye has been added for color (see 6.3.19). The coating shall completely cover the bottom of the squib case and may overlap the sides.

3.3 Performance and product characteristics.

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3.3.1 *Functioning time.* The squibs shall function within 0.003 second maximum, when subjected to a direct current (dc) firing current of 5 plus or minus 0.15 amperes at any temperature within the range of plus 71.11° to minus 58.33° plus or minus 1.11° C. (plus 160° to minus 65° plus or minus 2° F.) (see 4.4.3.3).

3.3.2 *Dielectric.* The resistance between the connected lead wires and the case shall be not less than 50 megohms when an alternating potential of 1000 volts is applied for one second (see 4.4.3.2).

3.3.3 *Electrostatic sensitivity.* The squib shall not function when exposed to a single discharge, between the connected lead wires and the case, from a 500 micromicrofarad (uuf) capacitor charged to a potential of 10,000 volts (see 4.4.3.7).

3.3.4 *Electrical firing sensitivity.* The squib shall not function when a current of 0.45 amperes flows through the bridgewire for 5 seconds minimum (see 4.4.3.5).

3.3.5 *Electrical continuity and resistance.* The electrical circuits of the squib between the lead wires shall be continuous and within the resistance limits of 0.70 ohms minimum and 1.30 ohms maximum plus 0.032 ohms per each double foot of lead wire (see 4.4.3.6).

3.3.6 *Environmental and service conditions.*

3.3.6.1 *Watertightness.* The squibs shall function within 0.005 second, after being submerged in water and exposed to conditions involving continuous changes in temperature, when subjected to a dc firing current of 5 plus or minus 0.15 amperes at normal room temperature (see 4.4.3.4).

3.3.6.2 *Shock resistance.* The squibs shall function, after exposure to shock conditions encountered during handling and storage, when subjected to a dc firing current of 5 plus or minus 0.15 amperes at normal room temperature. Except for equipment failure, any squib with a functioning time in excess of 0.004 or which fails to fire shall be considered a failure. In addition, the squibs shall meet the resistance and continuity requirements, both before and after shock resistance listing (see 4.4.3.1.1).

3.4 Workmanship. Uniformity of shapes, dimensions, and performance shall permit interchangeability of complete squib assemblies, without interference, bending, or misalignment. There shall be no cracks, breaks, chips, bends, dents, burrs, loose solder, loose attaching parts, loose electrical connections or any evidence or poor workmanship that is such that the squib is unsuitable for the purpose intended. The squib shall be clean and free of foreign material.

3.4.1 *Conditions of parts.* Inert parts as indicated on the applicable drawings shall be free of oil, grease and other foreign material prior to any assembly and the final assembly. Plastic or fiber parts shall, after fabrication and inspection, be protected from moisture or any other influence which might affect dimensional stability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall consist of one hundred (100) inert squibs parts and one hundred (100) loaded squibs. The preproduction sample shall be manufactured by the same method proposed for the manufacture of subsequent production of inert squib parts and the loaded squibs. The preproduction sample shall be subjected to all examination and tests specified herein. Unless otherwise specified, the

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Government will perform the examinations and tests for the preproduction sample.

4.3 Inspection provision.

4.3.1 *Lot size.* Each lot of squibs presented at one time for acceptance shall contain inert parts and loaded squibs, each of which is limited to one manufacturer.

4.3.2 *Sampling.*

4.3.2.1 *Sampling for examination.* Unless otherwise specified (see 6.2), sampling for examination shall be in accordance with MIL–STD–105, utilizing inspection levels and AQL's as specified in table I.

4.3.2.2 *Sampling for testing.* Unless otherwise specified (see 6.2), sampling for test of each production lot shall be in accordance with MIL–STD–105, utilizing inspection levels and AQL's as specified in table II.

4.3.3 *Examination.*

4.3.3.1 *Examination testing.* Unless otherwise specified, all examinations shall be conducted at room temperature. Examination testing of samples specified in 4.3.2.1 shall be conducted in accordance with Specification MIL–A–2550 for the purpose of determining compliance with the requirements specified in 3.2 and 3.4, except as covered elsewhere, using an AQL of 1.5.

4.3.3.2 *Inspection of preparation for delivery.* The preservation, packaging, packing and marking shall be examined to determine compliance with the requirements of section 5 and specifications and regulations referenced therein.

4.3.4 *Testing.* The contractor shall ascertain by certificate of analysis and testing that the squibs presented for acceptance conforms to the requirements for the following:

Characteristic

Lead wire (insulation material, see 3.2.1.1.3(b))
Composition of initiation charge (see 3.2.1.2.1.1)
Zirconium, type I (particle size, see 3.2.1.2.1.1.1)
Potassium perchlorate (ingredient percent, see 3.2.1.2.1.1.2)
Granulation I (average particle diameter, see 3.2.1.2.1.1.2.1)
Granulation II (particle size, see 3.2.1-.2.1.1.2.2)
Graphite powder (conductivity, purity and particle size, see 3.2.1.2.1.1.3)
Binder (ingredient percent, see 3.2.1.2-.1.1.4)
Loading initiation charge (weight of charge, correct, see 3.2.1.2.1.2)
Conductive coating of initiation charge (ingredients percent, see 3.2.1.2.1.3)
Composition of main charge (ingredient percent, see 3.2.1.2.2.1)
Zirconium, type II (purity and particle, size, see 3.2.1.2.2.1.1)
Aluminum, atomized (particle diameter, percent free aluminum, grease free, see 3.2.1.2.2.1.2)
Aluminum, flake (granulation, grease and oxide content, see 3.2.1.2.2.1.3)
Barium nitrate (ingredient percent and particle size, see 3.2.1.2.2.1.4)
Binder, type II (ingredient percent, see 3.2.1.2.2.1.5)
Main charge loading (weight of charge correct, see 3.2.1.2.2.2)

4.4 Tests. Unless otherwise specified, all tests shall be conducted at room temperature.

4.4.1 *Plug assembly test.*

4.4.1.1 *Finish and composition of plug.* At least one test shall be made for each mold per batch of material or whenever any major change is made in the curing procedure. Select ten sample plug assemblies and conduct an acetone extraction test in accordance with Method No. 7021 of Standard Fed–STD–406. An acetone extraction value exceeding 5 percent shall be cause for discontinuing production from the corresponding mold until such changes are made as are necessary to enable the plug to meet the requirements.

4.4.1.2 *Plug leakage.* Ten plugs per mold cavity per lot (minimum) and an additional 10 plugs shall be taken from each mold cavity from each batch of new material. These plugs

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shall be assembled in squib cases, with three drops of fluorescent penetrant placed in the bottom of the case prior to assembly. When assembled for this test, the plug assembly shall be located in approximately the same position relative to the case as is obtained in the completed squib. It is permissible to cut wire flush with plug or strip insulation away from plug for better observance of any penetration of the fluorescent penetrant. The assembly shall be so made that the contact surface between the plug and the case is not wet with the fluorescent penetrant. After assembling the plug and the case, and crimping the case, the test shall be conducted with the squibs in a vertical position with the lead wires down for a period of 72 hours. A daily inspection shall be made to determine the extent of leakage of fluorescent penetrant along the lead wires or through any portion of the plug. Developing powders and ultraviolet light shall be used as necessary in checking leakage. A record shall be kept of the daily inspection for leakage of the fluorescent penetrant.

4.4.1.2.1 *Evaluation of leak test results.* Except for leakage along contact surfaces, no assembly shall show leakage within 48 hours, and not more than one assembly for each mold cavity sample, ten shall show leakage within 72 hours. Leakage in excess of these limitations shall be cause for discontinuation of acceptance from the corresponding mold cavity until such changes are made as are necessary to enable the plug to meet the requirements.

4.4.1.3 *Plug and lead wire tightness.* A suitable test fixture shall be used to hold the plug so that the lead wires are maintained in a vertical position. Ten pounds, including weight of any necessary attaching device, shall be applied without shock, to each lead wire. Evidence of loosening of lead wires in the plug shall be cause for rejection.

4.4.1.4 *Dielectric test of plug.* A dielectric test on the samples specified in paragraph 4.3.2.1 shall be accomplished by the manufacturer, under the surveillance of the Government inspector, to determine conformance to the requirements of 3.2.1.1.1.1. Suitable current limiting resistors and resistance indicating device shall be provided in the test apparatus. Plugs having a resistance of less than 50 megohms shall be rejected (see table I).

4.4.2 *Bridgewire strength.* A suitable test fixture shall be used to subject each bridgewire to a load of 25 plus or minus 2 grams applied normal to the axis of the wire at its center. Assemblies which fail to meet this test shall be rejected.

4.4.3 *Acceptance tests of complete squib assembly.*

4.4.3.1 *Shock resistance.* A shock resistance test shall be conducted either according to MIL–STD–350 or the squib may be installed in a suitable test fixture and tested according to 4.4.3.1.1 (see table II).

4.4.3.1.1 *Shock resistance procedure.* The test fixture with squibs shall be dropped 20 successive times, top surface up, on 2 inch planking supported by a minimum of 3 inches of concrete, in such a manner that the test fixture lands squarely on its bottom surface. The drop height shall be a minimum of 5 feet. The fixture (with squibs) shall be dropped an additional 20 times in such a manner that the plate lands on one of the bottom edges, with the fixture making an angle of about 45 degrees with the planking. The angle drops shall be so made that each of the four bottom edges of the fixture receives five impacts. Following the total of 40 drops, the squibs shall meet the requirements of 3.3.6.2 (see table I).

4.4.3.2 *Dielectric test of squib.* A dielectric test shall be made according to 3.3.2 by the manufacturer under the cognizance of the Government inspector. Suitable current limiting resistors and resistance indicating device shall be provided in the test apparatus. Squibs found to have a resistance of less than 50 megohms shall be rejected. The test is nondestructive and good parts shall be returned to production. Sampling for Govern-

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ment inspection and lot disposition shall be in accordance with that shown in table I.

4.4.3.3 *Squib functioning time test.* The test is to ascertain that the squibs will function within the prescribed upper time limit of 0.003 second maximum, except for the shock resistance and watertightness tests when the function times shall be within 0.004 and 0.005 second maximum respectively (see table II).

4.4.3.3.1 *Squib functioning time test procedure.* The squibs shall be directed against a glass window; behind which is located a high vacuum photo tube having an S–1 or S–4 spectral response (see Figure 1 and Table II). An initial calibration of the test circuit will be made by connecting a precision (1 percent tolerance or less) one (1) ohm resistor across terminals A and B (fig. 1). The total resistance of the circuit, including the precision resistor, will be adjusted by varying the length of the resistance wire until a current of exactly 5 amperes flows. Switch S shall be closed and opened several times to assure constant current and voltage. Adjust circuit resistance if the current deviates from the setting of 5 amperes. Following the initial preparation of the test circuit the squib assemblies shall be fired in accordance with Figure 1 behind a suitable barricade. The timer or chronograph shall record the functioning delay time to an accuracy of 0.01 millisecond. In addition, occasional oscilliscope traces shall be photographed using 3000 film or faster, as a doublecheck of timer and also to assure that no spikes are present.

4.4.3.4 *Watertightness.* Samples selected in accordance with paragraph 4.3.2.1 shall be submerged in water and subjected to continuous changes of temperature in accordance with the procedure of 4.4.3.4.1 (see table II).

4.4.3.4.1 *Watertightness procedure.* The squib shall be submerged in water and subjected to continuous changes of temperature from 70° plus or minus 5°C. (158° plus or minus 9°F.), to that of tap water. A minimum temperature differential of 33.4°C. (60° F.), shall be maintained between the hot water and tap water temperatures. Each complete temperature change cycle shall be performed in the following manner: First, submerge to a minimum depth of six (6) inches in the 71.11°C. (160°F.), water for a minimum of five minutes and second, immediately submerge to a minimum depth of six (6) inches in the tap water for a minimum of five minutes. Each temperature cycle shall be repeated continuously for three complete cycles. The squib shall then be removed from the test chamber and test fired.

4.4.3.5 *Electric firing sensitivity.* Each squib shall be subjected to a current flow of 0.45 amperes, applied for a minimum of 5 seconds (see table II). Functioning shall not occur.

4.4.3.6 *Squib electrical continuity and resistance.* The contractor shall test each completed squib for electrical continuity and resistance using a resistance testing circuit which subjects the squib to a current less than 10 milliamperes. Resistance measurements shall be made with a suitable meter having an accuracy of plus or minus 0.01 ohm. Individual squibs with a resistance of less than 0.70 ohm or greater than 1.30 ohms plus 0.032 ohm for each double foot of lead wire shall be rejected.

4.4.3.7 *Electrostatic sensitivity.* An electrostatic sensitivity test shall be accomplished to determine compliance with 3.3.3. Each squib case shall be grounded to a common ground and the shunted lead wires shall be connected to a capacitor, which has previously been charged through a vacuum switch from a high voltage power supply. The capacitor shall be discharged between the squib case and shunted leads through a vacuum switch. No squib shall function with a 500 micromicrofarad capacitor charged to 10,000 volts. This test is nondestructive and good squibs shall be returned to production (see table I). See figure 2 for schematic of static sensitivity test circuit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preser-

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vation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning and drying.* Each squib shall be cleaned by process C-1 and dried by procedure D–1 or D–4 of MIL–P–116.

5.1.1.2 *Unit packaging.* A specified quantity (see 6.2) of squibs cleaned and dried as prescribed in 5.1.1.1 shall be immobilized in a snug-fitting container conforming to Specification PPP–B–566. The quantity of squibs for container shall not exceed 10 pounds. Immobilization shall be accomplished by using cushioning material conforming to Specification PPP–C–843. The packaged container shall then be inserted in a bag made from material conforming to class 1 of MIL–B–131 and heat sealed. The heat sealed package shall be inserted in a snug fitting outer container conforming to class 2 of Specification PPP–B–636. The outer container shall be closed by applying two strips of 2 inch width tape the length of the seams. Tape shall conform to Specification PPP–T–76. When the bottom seam is closed with adhesive or staples, a single piece of tape the length of the top seam is required. The method of preservation shall be 1A–14. of MIL–P–116. When intermediate packaging is specified, the outer container will be omitted and the preservation method will be 1A–15 of MIL–P–116. No overwrap of the unit package will be required.

5.1.1.3 *Intermediate packaging.* When specified (see 6.2), a number of unit packages packaged as prescribed in 5.1.1.2 shall be placed in a snug-fitting container conforming to class 2 of Specification PPP–B–636. Closure shall be accomplished by applying two (2) strips of 2 inch width tape the length of the seams. Tape shall conform to Specification PPP–T–76. When the bottom seam is closed with staples or adhesive, a single piece of tape the length of the top seam is required. When the intermediate container is the shipping container, closure shall conform to the requirements of Specification PPP–B–636, including the taping of the manufacturer's joint.

5.1.2 *Level C.*

5.1.2.1 *Cleaning and drying.* Cleaning and drying shall be as prescribed in 5.1.1.1.

5.1.2.2 *Unit packaging.* Unit packaging shall be as prescribed in 5.1.1.2 except that barrier material conforming to MIL–B–131 and the outer container conforming to class 2 of Specification PPP–B–636 shall be omitted. The preservation method shall be method III of MIL–P–116.

5.1.2.3 *Intermediate packaging.* Intermediate packaging shall be the same as Level A except that a container conforming to class 1 of Specification PPP–636 shall be used.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* A quantity of interior packages packaged as prescribed in 5.1 shall be immobilized in a snug-fitting container conforming to class 2 of Specification PPP–B–621. The maximum gross weight shall not exceed 150 pounds. Closure shall be in accordance with the appendix of Specification PPP–B–621.

5.2.2 *Level B.* Level B packing shall be the same as Level A except that containers conforming to class 1 of PPP–B–621 shall be used.

5.2.3 *Level C.* Squibs, electrical, packaged as prescribed in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the rules or regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, or Code of Federal Regulations 49 CFR 71–901, unit, intermediate packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The M3 Squib is intended for use in the M65 and XM57 Igniters and for the XM37E1 Spin Rocket Motor.

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6.2 Ordering data. Procurement documents shall include the following:

(a) Title, number and date of this specification.
(b) Whether or not preproduction sample is required (see 3.1).
(c) Applicable sampling plan and method of inspection, if other than specified herein.
(d) Selection of applicable levels of packaging and packing required.
(e) Special marking if required (see 5.3).
(f) Applicable unit and intermediate quantities (see 5.1.1.2).
(g) Whether or not an SOP is required (see 6.3).

6.3 Manufacturing processes and procedures. Manufacturing processes and procedures shall be left to the discretion of the contractor. However, details of manufacturing processes and the equipment to be used by the contractor shall be defined by a written standard which shall be submitted to the procuring activity for written approval prior to loading and assembly of the squib. Any alteration to or deviation from the approved standard shall be submitted in writing to the procuring activity, and written approval shall be obtained prior to use. Information contained in paragraphs 6.3.1 through 6.6 is considered extremely useful and reflects the current state-of-the-art relative to materials and processes for use and production of the M3 Squib.

6.3.1 *Lead wire spacing.* The distance between the lead wires shall be adjusted before soldering to give proper resistance to the completed bridgewire. The lead wires shall be bent together slightly, after soldering, in order to relieve tension on the bridgewire.

6.3.2 Potassium perchlorate of "Certified Reagent Grade" available from Fisher Scientific Company, New York, New York, has been found satisfactory.

6.3.3 "Ultra Purity Graphite, type UCP–2 (–325 mesh)" manufactured by United Carbon Products Company, Bay City, Michigan, has been found satisfactory.

6.3.4 The binder "VYLF" manufactured by the Bakelite Division of Union Carbon and Carbide Company, New York, N. Y. has been found satisfactory.

6.3.5 *Preparation of initiation charge.* The ingredients are finely ground and uniformly mixed using normal butyl acetate to wet the mixture. Dissolve the binder in normal butyl acetate before it is added to the mixture.

6.3.6 *Loading initiaton charge.* The initiation charge completely covers the bridgewire and is free of voids, cracks, or fissures. The initiation charge covers a minimum of area on the exterior surfaces of the lead posts and should be at least 0.025 inch below the top of the spacer. Two or more applications of the initiation mixture are required. The first application coats the bridgewire with a thin film, and the successive applications build up the bead of initiation mixture to the required weight of 0.012 plus or minus 0.003 grams. All excess initiation mixture is cleaned from the assembly by using muslin wet with normal butyl acetate. After loading, dry initiation charge not less than 8 hours at a temperature between 43.33° and 76.67° Centigrade (C) (110° and 170° Fahrenheit (F)).

6.3.7 Dissolve the binder in normal butyl acetate and mix with the graphite. Apply the coating by brushing to form a thick, continuous conductive shield over the entire initiation charge. Coat the upper portion of the lead post and leave some part of the lower portion exposed. Do not allow the conductive coating to run down the side of the squib plug or appear on any surfaces except the 0.191 inch diameter face of the initiation assembly shown on Drawing 8030864.

6.3.8 *Zirconium, type II.* The zirconium purity and particle size and critical factors in meeting performance requirements. Zirconium, type II, is produced by grinding solid fuzed material. Granular zirconium manufactured by Foote Mineral Company, Philadelphia, Pennsylvania.

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6.3.9 *Aluminum, atomized.* Type I-511 manufactured by Reynolds Metal Company, Louisville, Kentucky, has been found satisfactory.

6.3.10 *Aluminum, flake.* Alcoa 606 manufactured by Aluminum Company of America, Pittsburgh, Pennsylvania, has been found satisfactory.

6.3.11 *Barium nitrate.* Barium nitrate of "Analytical Reagent Grade" manufactured by Mallinckrodt Chemical Works, St. Louis, Missouri, has been found satisfactory.

6.3.12 *Binder, type II.* "XYHL" manufactured by Bakelite Division of Union Carbon and Carbide Company, New York, New York, has been found satisfactory for use as a type II binder.

6.3.13 *Main charge preparation.* The ingredients of the main charge should be mixed wet with 95 percent ethyl alcohol after having been previously ground and weighed dry. Dissolve the binder in ethyl alcohol before it is added to the mixture.

6.3.14 *Main charge loading.* After mixing, butter approximately 90 milligrams of the main charge into the spacer around the initiation charge, stir the main charge mixture frequently to prevent settling. Load the main charge flush with the top of the spacer and remove all excess material at the top, or on the sides of the spacer. The main charge should be buttered in such a manner as to produce a minimum of voids or fissures.

6.3.15 *Squib assembly.* After loading the initiation charge and applying the conductive coating, cement the spacer ring to the squib plug in such a manner as to form a dielectric seal. After the cement is dry, load the main charge and dry the squibs in an oven. Then insert the insulating liner between the squib subassembly and case, and crimp the case in place. Perform the mating of the squib charge and the case in a time interval not to exceed 2 hours from the time of removal of the main charge assembly from the oven or desiccator. Perform the crimping in such a manner as to prevent scarring or undue strain in the bottom of the case.

6.3.16 *Cementing spacer ring to squib plug.* Cement the spacer ring to the squib plug in such a manner as to form a dielectric seal. Do not permit the cement to contact the initiation charge or the lead posts, or to protrude on the exterior of the plug in such a manner as to interfere with the mating of the case and squib charge. Apply the spacer ring cement uniformly to the squib plug and spacer ring. The following cements have been found satisfactory for cementing the plug and spacer ring:

General Electric Company Pittsfield, Mass.	— Glyptal 1201 Red Enamel
Minnesota Platon Corporation Pipestone, Minn.	— Poly-up Platon,
Bakelite Division of Union Carbon and Carbide Company, New York, N. Y.	— Binder, type I, 30 *percent solution* in normal butyl acetate.

6.3.17 *Insulation liner insertion of.* The insulating liner disc shall be inserted between the case and the main charge assembly. The disc overlap over the spacer shall be not greater than 0.125 and not less than 0.093 inch. A minute amount of solvent-free adhesive or grease may be used to hold the disc in place before assembly. The thickness and dielectric strength of the disc shall not be changed or damaged.

6.3.18 *Insulating liner.* The insulating liner shall be a polytrifluorochoroethylene film which is water repellant, chemically resistant, and has a high dielectric strength. The film shall have a low electrical fault count and shall be of a type suitable for use

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as a dielectric in high temperature applications.

6.3.19 *Case coating dye.* Dyes manufactured by National Analine Division of Allied Chemical and Dye Corporation has been found satisfactory.

6.4 Safety regulations. The loading, assembly, and handling of the squibs and related explosive components thereof, covered by this specification, involve hazardous operations and therefore require the observance of suitable safety precautions. Use of this specification shall not be construed to relieve the contractor or the manufacturer of responsibility for the safety of his operations.

6.5 Handling. Personnel handling the loaded squibs and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.6. Precautions peculiar to zirconium. Two types of zirconium are used in the M3 squib. Type I zirconium as used in the initiator charge is finely divided powder produced by reduction of zirconium dioxide with an alkaline earth metal in a closed bomb. Type I zirconium powder is easily ignited by static electricity or simple friction when dispersed. The ignition point of dry powder is between 181.6° and 195°C. (359° and 383°F.). If the powder is ignited in bulk, it will glow or burn slowly. On the other hand, if the powder is dispersed in the air and ignition occurs, an explosion may result. In the form of a wet paste the powder is relatively safe. However, if the wet paste is exposed to fire, it may burn more violently than dry powder. Type I zirconium must be stored and shipped in a water wet condition. The powder should always be handled in small batches (100 grams or less) and be kept wet through as many of the operations as possible. Any spillage should be cleaned up immediately and completely. The water may be removed from the zirconium by washing with absolute methyl alcohol and then drying at room temperature. Washing and drying should be carried out on small quantities, less than 100 grams. The water wet paste may be slurried and filtered on a Buchner funnel using a No. 41 Whatman filter paper. After the water has been removed, additions of alcohol can be made without disturbing the cake of zirconium. Approximately 12 hours of air drying is required to completely dry the powder. The powder should be weighed, and added to the other ingredients of the initiation charge wet with normal butyl acetate. Type II zirconium, used in the main charge is produced by grinding fused solid material and is relatively safe to handle. The granular zirconium is insensitive to ignition by friction or static electricity relative to Type I zirconium. Type II zirconium may be shipped and handled dry using precautions normally used with ground magnesium.

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TABLE I. *Examination tests for M3 squib*

Reference paragraphs	Type of test description	Sampling plan	AQL
3.2.1.1.1.1 4.4.1.4.	Dielectric test of plug (50 megohms resistance at 1000 volts ac).	MIL–STD–105 *Inspection* Level II.	0.065
4.4.3.2 3.3.2	Dielectric test of Squib (50 megohms resistance at 1,000 *volts* ac).	MIL–STD–105 Inspection Level II or 100% Inspection.	0.065 or reject individual defectives.
4.4.3.7 3.3.3	Electrostatic sensitivity, no functioning at 10,000 volts, 500 uuf.	MIL–STD–105 Inspection Level II or 100% *Inspection*.	0.025 or reject individual defectives.

TABLE II. *Inspection tests for M3 squib*

Reference paragraphs	Type of test description	Sampling plan	AQL
4.4.3.1 3.3.6.2	Shock test–5 ft. drop or jolt test MIL–STD–350 (0.004 seconds functioning time).	MIL–STD–105 Single Sampling Plan Table IV-A Code Letter J	1.0%
4.4.3.5 3.3.4	*Firing* Sensitivity (0.45 amp dc).	MIL–STD–105 Plan Table III-B Inspection Level L-6	0.065
4.4.3.3	Squib functioning (0.003 second upper limit with 5.0 plus or minus 0.15 amps at 24 volts dc).	MIL–STD–105 Single Sampling Plan Table IV-A Code Letter J	1.0%
4.4.3.4 3.3.6.1	Watertightness—Fire sample at the end of cycling (0.005 second upper limit with 5 plus or minus 0.15 amps dc).	MIL–STD–105 Single Sampling Plan Table IV-A Code Letter J	2.5%

A dud shall be defined as a failure to fire.
When 2 or more duds (see 3.3) occur on watertightness test in samples specified in table II, the low shall be rejected.
When 2 or more duds (see 3.3) occur on jolt test in samples specified in table II, the lot shall be rejected.
If 1 or more duds occur (see 3.3) on squib functioning tests in samples specified in table II, the lot *shall* be *rejected*.

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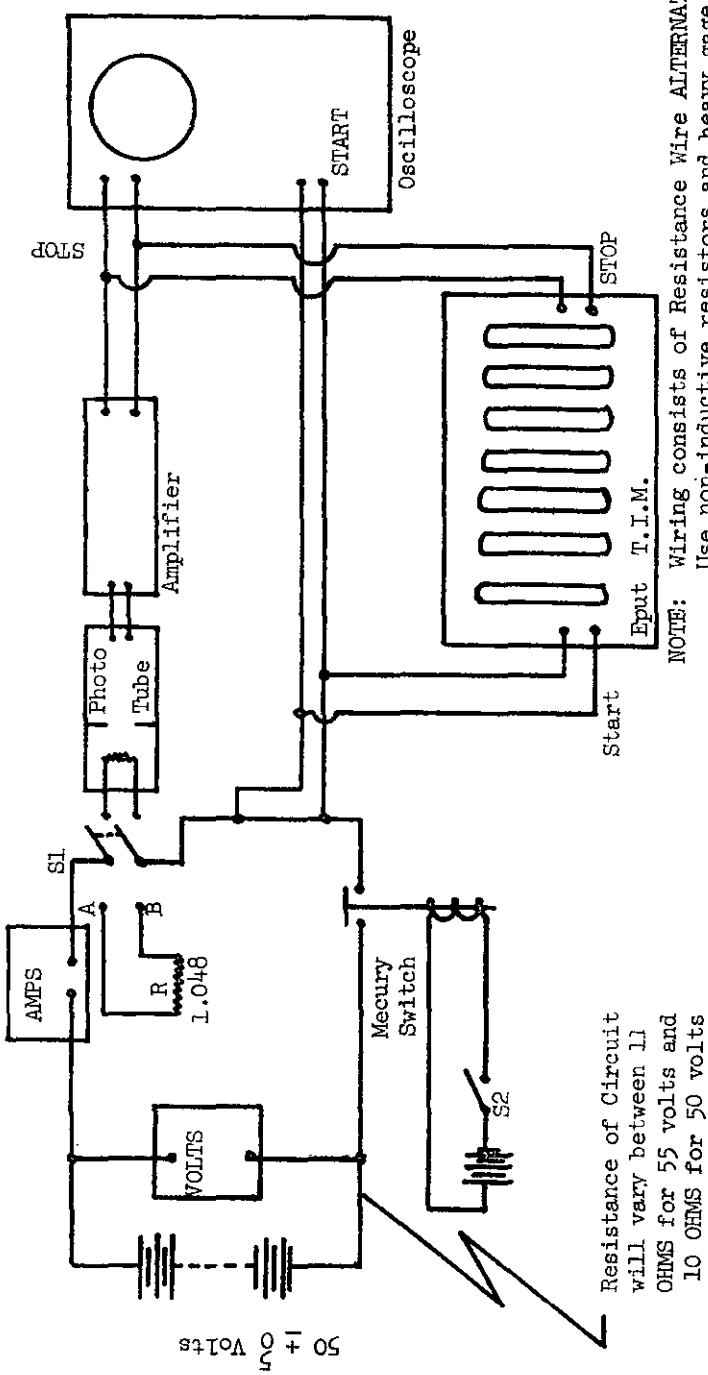
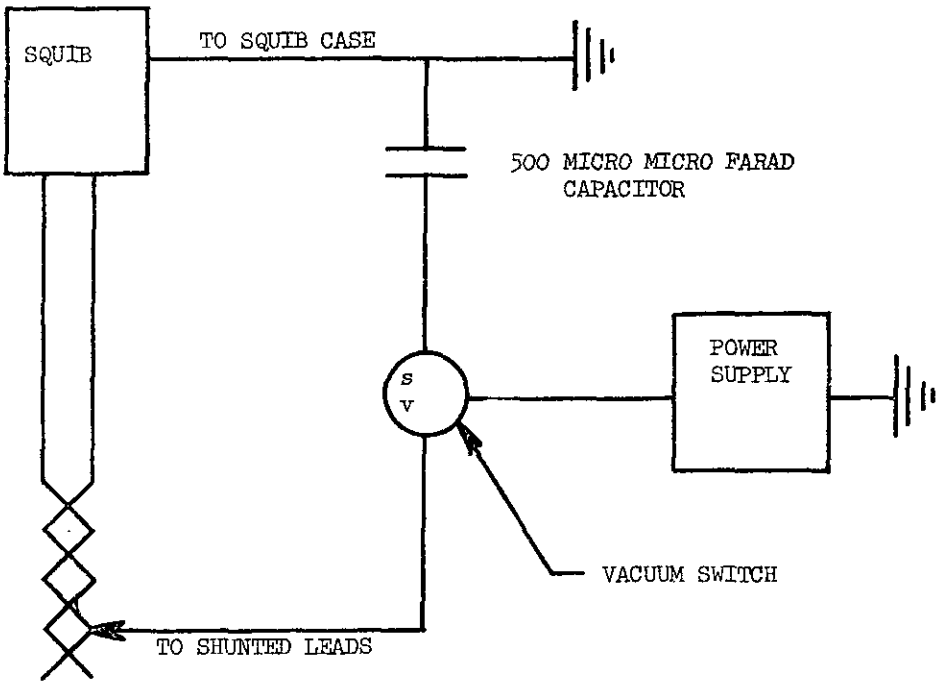


FIGURE 1.

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SCHEMATIC OF STATIC SENSITIVITY TEST CIRCUIT

FIGURE 2.

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